

BIO504T OBJECTIVES

FINAL TERM-2024- 100%

BIOCHEMISTRY I

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OBJECTIVES

- ❖ Guanine represents by symbol G
- ❖ Standard amino acid are 20
- ❖ tRNA in cell are 20%.
- ❖ The deleterious effect of trans fat occurs at intake of 2 to 7 g\ day.
- ❖ At 5' end there is free phosphate present.
- ❖ Three bonds present in which two bases.. G and C
- ❖ By which attraction purine and pyrimidine are bound: hydrogen bonds
- ❖ Molecular weight of water: 18
- ❖ Uric acid is the catabolic end product of purines in human beings.
- ❖ Cholesterol synthesis occurs in: liver
- ❖ Salt of which substance are water soluble: ...potassium (K)
- ❖ Medium chain fatty acid contains what number of fatty acids: (6-12 C)
- ❖ Homopolysaccharides (homoglycans): Polymers of same monosaccharide units
e.g. starch, glycogen, inulin, cellulose, dextrans, dextrans
- ❖ Transform of maleic acid... Fumaric
- ❖ Crystalline glucose is .. α -D - glucopyranose
- ❖ The net charge on alanine in acidic solution (pH less than 2) is.
Positive
- ❖ the pK₂ of glycine is -----?? 2.34
- ❖ Keratin is found..... in hair, nail, hoofs
- ❖ β Sheets are a second type of regular secondary structure that maximizes hydrogen bonding between the peptide backbones.
- ❖ OxyHemoglobin Dissociation Curve Bohr effect decreased affinity of haemoglobin for --
--gas caused by an increase of carbon dioxide pH etc.
Oxygen
- ❖ Lungs have pH around 7.40, while tissues have pH around 7.35-7.45.
- ❖ Glutamate is replaced with..... valine
- ❖ Medium chain of fatty acid contain..... (6-12 C)
- ❖ The melting point of fatty acid depends upon chain length and (degree of unsaturation)
- ❖ The common name of w6 is..... linoleic acid
- ❖ No. Of carbon atom in stearic acid? (18:0)

- ❖ The deleterious effect of trans fats occur at intake of2 to 7 g / day.
- ❖ which one is halogen... bromine.
- ❖ Adenine bind with..... thymine and uracil.
- ❖ The small organic molecule that assist enzyme function..... coenzyme.
- ❖ stearic acid.....18.
- ❖ The mammalian ribosome consists of a large subunit (60S) and a small subunit (40S).
- ❖ These enzyme catalyzes condensation reaction joining two molecules by forming C-O ,C-S,C-N and C-C bond ligase.
- ❖ Adenine bind with: Thymine (in DNA) or Uracil (in RNA)
- ❖ The mammalian ribosomes contains two major nucleoproteins subunits a large subunit (60S) and a small subunit (40S)
- ❖ The bond present between the double helix: hydrogen bonds.
- ❖ Double bond is present in: Thymine and adenine(A-T)
- ❖ pH is controlled by: buffer solution
- ❖ types of enzymes: 6
- ❖ Keratin is found: in hair, nail, hoofs.
- ❖ Saturated fatty acids have double bonds: Zero
- ❖ pH of glutamate: pH 7.4
- ❖ Medium Chain Fatty Acids: (6-12 C)
- ❖ Which of the following is pyrimidine: cytosine, thymine, and uracil.
- ❖ Serine + palmitate = (Sphingolipids)
- ❖ The sign of enthalpy is (ΔH)
- ❖ Measure of randomness system is (Entropy)
- ❖ Negative charge on DNA is due to (phosphate group)
- ❖ At a five end of RNA, there is a due to free (Phosphate group)
- ❖ The bond between DNA and RNA----? Phosphodiester Bond.
- ❖ The - strand is called Template strand while 3 to 5 direction
- ❖ Transform of malic acid is called..... Fumaric
- ❖ Num. Of Stereoisomers for a molecule containing only one chiral carbon atom..... 2
- ❖ The simplest aldose, glyceraldehyde contain one chiral center and therefore has..... two different optical isomers
- ❖ The bond exists between the carbonyl group and oxygen of the hydroxyl group in ring structure of glucose is..... Covalent
- ❖ L- Rhamnose is found in polysaccharides of in plant polysaccharide
- ❖ The action of acid on carbohydrates form the basis of certain test of determine? Benedict's test
- ❖ Among the following table sugar is termed as..... Sucrose
- ❖ An average protein will not be denatured by..... physiological conditions or mild heat
- ❖ Major hemoglobin in adults is.....Hemoglobin A (HbA)
- ❖ In sickle cell anemia, Glutamate is replaced with.....Valine
- ❖ Medium chain fatty acid contain..... (6-12 C)
- ❖ Arachidonic acid can be synthesised by..... both Linoleic acid and alpha-Linolenic acid.
- ❖ The num. Of carbon atom in palmitic acid is 16:0
- ❖ Triacylglycerols areare nonpolar, hydrophobic molecules, essentially insoluble in water
- ❖ Rancidity is caused due to..... oxidative reactions

- ❖ The chief constituent of bee wax is..... **Triacontanol ,also known as melissyl alcohol or myricyl alcohol.**
- ❖ saturated lipids increase __ level in blood? **Increase**
- ❖ Ethanolamine is..... **primary alcohol and a primary amine**
- ❖ Cholesterol synthesis occur in**liver**
- ❖ The type of sugar in DNA is.....**deoxyribose**
- ❖ Adenine bind to..... **Thymine (in DNA) or Uracil (in RNA)**
- ❖ The pentose sugar of RNA is..... **ribose**
- ❖ The Lineweaver-Burk plot is diagnostic for.....**enzyme kinetics**
- ❖ PH of buffer solution depends upon concentration of **both the acidic and basic components.**
- ❖ Which of the following common drug is not a specific enzyme inhibitor.....
Acetaminophen (paracetamol)
- ❖ The type of RNA acts as machinery for the synthesis of protein is..... **m RNA**
- ❖ The higher the content of GC base pairs, the **higher** the melting point of the DNA.
- ❖ Hemoglobin is excessively found in _? **RED Blood cells**
- ❖ Template strands are read in thedirection?: **3' to 5'**
- ❖ Reversible inhibitor the value of V_{max} ? : **decrease**
- ❖ Phosphatidyl choline is formed by-----? **sphingosine and choline**
- ❖ Platelet aggregation is the culminating step in the cardiovascular diseases due to?
atherosclerosis
- ❖ The free group attached at the 3' end of nucleotides is the? **Hydroxyl (OH) group**
- ❖ Hemoglobin is a.....**(Tetramer)?**
- ❖ Glucuronic acid is formed by the oxidation of glucose at.....?: **C-1**
- ❖ These enzymes catalyze condensation reactions joining two molecules by forming C-O, C-S, C-N and C-C bonds.....? **ligases**
- ❖ DNA contains four bases that are?: **adenine (A), thymine (T), cytosine (C), and guanine (G).**
- ❖ The adjustment in the BPG level has on the binding of oxygen in the lungs?: (only a small effect)
- ❖ **Homopolysaccharides (homoglycans):** Polymers of same monosaccharide units e.g. starch, glycogen, inulin, cellulose, dextrans, dextrans
- ❖ Transform of maleic acid... **Fumaric**
- ❖ Type of carbohydrates that can undergo further hydrolysis into simple form.....**Monosaccharide**
- ❖ sucrose is **Non reducing sugar** .
- ❖ A structure homopolysaccharide is**ALL(starch.glycogen . cellulose)**.
- ❖ net charge of alanine **+1** .
- ❖ hydrogen group of Several sites attached for oligosaccharides chain in protein
serine and threonine
- ❖ sickle cell anemia that replaces the position of 6 beta chain of hemoglobin is
glutamate
- ❖ function of protein depends on **Sequence of amino acids**
- ❖ domains are connected through **Covalent bond(not sure)**.
- ❖ Two domains are connected through a **simpler structure like a loop**
- ❖ measure of the random or disordered system **entropy**
- ❖ Enthalpy denoted by **ΔH**
- ❖ The family chaperones that bind short sequence of hydrophobic Amino acid in newly synthesized Amino acid....**HSP 70.**
- ❖ adjustment in BPG level has what effect on the binding of oxygen in lungs **slow** .
- ❖ medium fatty acid chain.....**6-12 carbon** .

- ❖ the loose packing of unsaturated fatty acid is due to cis isomers of unsaturated *fatty acid* salt are water soluble Sodium and potassium
- ❖ common name of 6 mega linoleic
- ❖ number of carbon atoms in palmitic acid is 16
- ❖ in triglycerol the fatty acid are jointed to glycerol..... Ester .
- ❖ which one following is halogen..... Bromine .
- ❖ chief constituent bees wax. Myricyl palmital .
- ❖ when the cholesterol is taken up by HDL it is immediately estrified and become..... Hydrophobic
- ❖ phosphotidycholine is formed from..... phosphatic and choline
- ❖ carbohydrates and lipids together form ... glycosphingolipids.
- ❖ cholesterol synthesize occur in..... All.
- ❖ double bond present in.... Adenine and thymine
- ❖ enzyme are divide in class Six major
- ❖ ph2 actively pepsin active.
- ❖ Total circular RNA . tRNA compose roughly 20 percent.
- ❖ adenine bend to..... Thymine.
- ❖ Genertic material for some animal and plant virus es All (DNA. RNA. Plasmid)
- ❖ most important to michoelist mental kinetic.... Regulatory enzymes
- ❖ bond water usually..... Hydrogen bonds
- ❖ strong fully ionized solution are... All(HCl. H2so4. Hno3).
- ❖ transition state of catalyzed reaction than that of unsaturated reaction.... Low energy
- ❖ at physicalogic ph which form nitrogen base.... Keto , amine
- ❖ The backbone of mono sacchrde carbohydrates unbranched carbon chains
- ❖ No.of carbon in palmitic acid... 16:0
- ❖ Machinery of making proteins..... m RNA
- ❖ Adenine make bond with... Thymine
- ❖ The triple bond between.... guanine and cytosine.
- ❖ The iodine number will.increase by.... increase in the number of double bonds or unsaturation in the fatty acid chains of a fat or oil.
- ❖ The structure of the proteins drawn on paper..is mai rule k pta ni.law k.options thy...
- ❖ Arcahnoidc acid is made from.... Linoleic acid and alpha-Linolenic acid
- ❖ The sinlge letter to represent the Aspartate: D
- ❖ Acid jitna zyada strong ho ga ph value utni kya ho hia...low..
- ❖ pKa of the -COOH group in the range of 1.8 to 2.4
- ❖ pKa of the -NH3+ group in the range of 8.8 to 11
- ❖ purines include: hypoxanthine and xanthine
- ❖ Purine bases: Adenine (A) and Guanine (G)
- ❖ The deleterious effects of trans fats occur at intakes of 2 to 7g/Day
- ❖ The mammalian ribosomes contains two major nucleoproteins subunits (Large 60S smaller 40S)
- ❖ Serin +palmitate = Sphingosine
- ❖ Higher GC base pairs increases Tm, because with three hydrogen bonds, require More heat energy
- ❖ Acid dissociation constant are designated as PKa
- ❖ The bond present between the double helix hydrogen bonds.
- ❖ Which one of the following is made up of two glucose maltose
- ❖ Music maltose comprises two glucose molecule that join together by..... Glycoside linkage (alpha 1 to4)
- ❖ Hemoglobin is excessively found in RED Blood cells

- ❖ Titration curve of Glycine known as Diprotic form
- ❖ The sign of enthalpy is ΔH
- ❖ In a double-stranded nucleic acid, cytosine typically base pairs with Guanine (C-G).
- ❖ Waxes are insoluble in Water
- ❖ Measure of randomness system is Entropy
- ❖ These enzymes catalyze condensation reaction joining two molecules by forming C*O C*s C*N C*C bonds ligases.
- ❖ Rancidity occurs by Oxidation
- ❖ Globosides are highly abundant in RBCs
- ❖ (very low density lipoproteins)
- ❖ VLDLs composed predominantly of TAGs
- ❖ Ethanolamine is a primary alcohol and a primary amine.
- ❖ Negative charge on DNA is due to Phosphate
- ❖ RNA refers to Ribonucleic Acid.
- ❖ At the 5' end of RNA, there is a free Phosphate group.
- ❖ The bond between DNA and RNA is a Phosphodiester Bond.
- ❖ Which one is halogen _____
- ❖ Halogen refers to elements like Fluorine, Chlorine, Bromine, Iodine, and Astatine.
- ❖ No. of carbon in steric acid is 18
- ❖ Ketokinase is involved in the metabolism of Ketone Bodies, not glucose.
- ❖ A motif also called a super secondary structure
- ❖ The smaller RNA is tRNA (Transfer RNA).
- ❖ Fruits contain Fructose.
- ❖ The alpha helix is present in the Secondary Structure of proteins.
- ❖ Maltose sunflower-shaped
- ❖ Iodine number of cotton seed oil varies from 103 to 111.
- ❖ That of olive oil from 79 to 88,
- ❖ And that of linseed oil from 175 to 202
- ❖ A commercial lot of olive oil which has iodine number higher than 88 might have been adulterated with cotton seed oil
- ❖ The enzyme for lactose to lactic acid conversion is Lactase.
- ❖ D-glucose has a hydroxyl (OH) group at the 5th carbon and an aldehyde group at the 1st carbon.
- ❖ phosphatidylcholine only in that ethanolamine or serine
- ❖ Ethanolamine has the structure NH₂-CH₂-CH₂-OH.
- ❖ The complex folding of chromosomes is referred to as Chromatin.
- ❖ The complex folding of large chromosomes within eukaryotic chromatin and bacterial nucleoids is generally considered tertiary structure
- ❖ acid dissociation constantpKa
- ❖ Weak Acids and Bases have Characteristic Acid Dissociation Constants
- ❖ Acid dissociation constant is represented as pKa.
- ❖ A nucleoside is formed by combining a nitrogenous base with a sugar molecule.
- ❖ Nucleoside is.....guanine + ribose
- ❖ Double bond is present in.....Thymine and adenine
- ❖ pH is controlled by.....buffer solution
- ❖ Standard amino acids are the 20 common amino acids found in proteins.
- ❖ Fatty acids contain a hydrophobic hydrocarbon tail and a carboxyl group.
- ❖ A fatty acid consists of a hydrophobic hydrocarbon chain with a terminal carboxyl group that has a pKa of about 4.8
- ❖ Complex structures of proteins are termed Tertiary Structures.

- ❖ Factors are determined by parameters like **Km, VMAX, or curve plots.**
- ❖ tRNA constitutes about **15-20%** of the total **RNA content** in a cell.
- ❖ Thymine and uracil differ by the presence of a **methyl group in thymine.**
- ❖ DNA denaturation is often monitored **at 260 nm.**
- ❖ Water is essential for the **living organisms.**
- ❖ Purines (adenine and guanine) have **NH2 groups.**
- ❖ The pKa value of glutamate is around **4.07.**
- ❖ **Fumaric** acid is converted to **Malic acid.**
- ❖ The Trans form of maleic acid is called **Fumaric**.
- ❖ **mRNA** constitutes the coding sequence for **proteins.**
- ❖ **Uric acid** is the catabolic end product of purines in human beings.
- ❖ Purines are converted into **uric acid.**
- ❖ In sickle cell anemias glutamate is replaced by **valine**
- ❖ VLDL test gives information about **lipid metabolism.**
- ❖ Aromatic non polar compounds _ **Phenylalanine – Tyrosine – Tryptophan**
- ❖ types of enzymes _ **6.**
- ❖ enzymes are divided into **six** major classes
- ❖ Cytosine forms pair with **Guanine.**
- ❖ Two Bonds between **A and T**
- ❖ Three between **G and C**
- ❖ Waxes are in water **Insoluble**
- ❖ **Purines** pyrimidine are **hydrophobic** in nature's
- ❖ Template strands are read in the **3' to 5' direction.**
- ❖ **Competitive** inhibitors resemble the substrate and compete for binding to the active site of the enzyme.
- ❖ **Noncompetitive** inhibitors do not bind at the active site
- ❖ **Competitive** inhibition shows a characteristic Lineweaver-Burke plot
- ❖ If the non-protein moiety is a metal ion such as Zn²⁺ or Fe²⁺ it is called a **cofactor.**
- ❖ **Cofactors** are inorganic ions or non-protein molecules necessary for enzyme activity.
- ❖ If it is a complex organic molecule or metallo-organic compound it is termed a **coenzyme.**
- ❖ **Coenzymes** are small organic molecules that assist enzyme function.
- ❖ Iron is a **transition** metal
- ❖ an iron atom has a positive **two charge** (ferrous ion) instead of **three charge (ferric ion)**
- ❖ Iron serves as a cofactor in **heme-containing proteins.**
- ❖ The names of pyrimidines are **cytosine, thymine, and uracil.**
- ❖ The **order of reactions** refers to the sequence of chemical events in a process.
- ❖ **Carbohydrates** can be classified into monosaccharides, disaccharides, and polysaccharides.
- ❖ Aromatic amino acids are **phenylalanine, tryptophan, and tyrosine.**
- ❖ **Halogens** include fluorine, chlorine, bromine, iodine, and astatine.
- ❖ An amino acid that is negatively charged at physiological pH is **Aspartate (Asp) & Serine**
- ❖ **Purines pair** adenine with thymine (A-T) and guanine with cytosine (G-C).
- ❖ The first leukotriene discovered was Leukotriene C. **(WBCs Leucocytes)**
- ❖ Leukotrienes your name so because they were first discovered in **(white blood cells)**
- ❖ **INTRONS** is removed during RNA maturation?
- ❖ activators consist how many domains? **Multiple**
- ❖ what charge is on DNA? **NEGATIVE**

- ❖ The mixture of DNA and proteins is called Chromatin.
- ❖ The point mutation in the DNA sequence result in the replacement of; Glutamate residue with Valine residue
- ❖ Protein mutation involves changes in base pairs.
- ❖ RAN refers to Ribonucleic Acid
- ❖ The pH of pepsin is around 2
- ❖ Water is the most abundant substance in living systems, making up 70% or more of the weight of most organisms
- ❖ Enzyme inhibitors are molecular agents that interfere
- ❖ with catalysis, slowing or halting enzymatic reactions.
- ❖ Any substance that can diminish the velocity of an enzyme-catalyzed reaction is called an inhibitor.
- ❖ which of the following enzyme inhibitors __ aspirin
- ❖ The pKa value of lactic acid is around 3.9.
- ❖ Substrate binds to the active site of the enzyme.
- ❖ TAGs containing saturated fatty acids are solid at room temperature such as butter whereas
- ❖ TAGs containing unsaturated fatty acids are liquid at room temperature such as olive oil. (Oleic acid, 18:1,9)
- ❖ Fats rich in saturated fatty acids have low iodine numbers,
- ❖ while fats rich in unsaturated fatty acids have high iodine numbers
- ❖ Saturated fatty acids contain no double bonds.
- ❖ Double bonds in unsaturated fatty acids 8 the mammalian ribosomes contain which two major nucleoprotein subunit.
- ❖ tRNA is the smallest of the three major species of RNA (4S).
- ❖ The higher the content of GC base pairs, the higher the melting point of the DNA.
- ❖ which of the following only present in RNA __ Uracil
- ❖ Both 5'-end and 3'-end are free. at 5'-end there is a free phosphate. at 3'-end there is a free OH
- ❖ The backbones of both DNA and RNA are hydrophilic.
- ❖ The ribosyl moieties are attached to the nucleobases by N-glycosidic bonds
- ❖ N-glycosidic bonds, have Nitrogen atom instead of oxygen linking the two residues.
- ❖ Sugars are linked to the heterocycle by a β -N-glycosidic bond, almost always to the N-1 of a pyrimidine & N-9 of a purine 15
- ❖ Simple triglycerides: Fatty acids connected to glycerol are of the same type e.g., tripalmitin.
- ❖ The Mediterranean diet is associated with decreased serum total cholesterol
- ❖ The deleterious effects of trans fats occur at intakes of 2 to 7 g/day
- ❖ waxes are insoluble in __ Water
- ❖ Glycerol is synthesized from Dihydroxyacetone Phosphate (an intermediate of the glycolytic pathway)
- ❖ Glycerol can form dihydroxyacetone phosphate which can convert into glucose or glyceraldehyde phosphate to enter into gluconeogenic pathway.
- ❖ The hsp70 family of chaperones binds short sequences of hydrophobic amino acids in newly synthesized polypeptides.
- ❖ Reducing disaccharides also give osazones Maltose sunflower-shaped
- ❖ Sunflower osazones are Maltose in nature.
- ❖ Pepsin is active at pH 2.
- ❖ The pka value of acetic acid is 4.76.
- ❖ The beets sheets form the core of globular proteins.

- ❖ Leukotrienes are named because they were first identified in white blood cells .
- ❖ About 78% of the amino acid residues in the myoglobin are found in α helices with bends in between.
- ❖ Ethanolamine is a primary alcohol and a primary amine.
- ❖ Galactolipids constitute about 70% to 80 % of plant membrane lipids.
- ❖ mRNA makes 2-5% of cell.
- ❖ RNA referred as Ribonucleic Acid